

CardioQ Heart Score — Methodology Note

Author: Swayam (IIT Bombay) | **Role under evaluation:** Product Development Intern, CardioQ.ai **Submitted to:** Capt. Indra Kumar Jha, Founder & CEO, Ziffytech Digital Healthcare **Build window:** 48-hour AI-augmented sprint

1. Problem statement

Build a working prototype of a CardioQ Heart Score engine: a single application that takes a patient’s clinical inputs, runs three globally validated risk algorithms (Framingham 2008, ESC SCORE2 2021, AHA PREVENT 2023) in parallel, and returns one unified, explainable Heart Score on a 0-100 scale, with a risk band, top contributing factors, and an exportable report.

This document describes the methodological choices behind the engine and is intended to be read alongside the source code in this repository.

2. The three sub-models

Model	Year	Outcome predicted	Validation age range	Inputs mod
Framingham General CVD	2008	Composite CVD: coronary death, MI, coronary insufficiency, angina, ischemic/hemorrhagic stroke, TIA, PAD, HF	30-74	Age, sex, total cholesterol, HDL, SBP (treated/untreated), smoker, diabetic
SCORE2 / SCORE2-OP	2021	Fatal + non-fatal CVD (MI, stroke, CV death)	40-69 / 70-89	Age, sex, systolic blood pressure, total cholesterol, HDL, regional calibration
AHA PREVENT (base)	2023	10-yr ASCVD (MI + stroke)	30-79	Age, sex, total cholesterol, HDL, SBP, statin use, diabetic, smoker, BMI

The three models were selected by the assignment brief; they are the most recent and most clinically endorsed in their respective regulatory jurisdictions (US, EU, AHA-revised US).

A note on outcome heterogeneity that we make explicit rather than hide: Framingham predicts a broader CVD composite (it includes heart failure and TIA, which the others don’t). SCORE2 includes both fatal and non-fatal CVD mortality. PREVENT’s primary 10-year output is ASCVD-only (the heart-failure component is modelled separately).

When we average the three, we are *approximately* combining three different predictions of “the same broad underlying risk”, which is defensible at this prototype stage but a known limitation for clinical use.

3. Coefficient sourcing and verification

A key concern was that AI recall of recent equation coefficients can be unreliable. The assignment brief instructed: “*we expect you to verify by reading the paper, not just trust Claude’s recall.*” We followed this strictly:

- **Framingham 2008 coefficients** were taken directly from the [Framingham Heart Study’s official risk-function page](#), which publishes the regression coefficients, baseline survival, and the centring constant. We then verified the implementation by hand-tracing one case (61F, TC 180, HDL 47, SBP 124, no other risks): $LP = 26.4390$, $LP - \mu = 0.2459$, $\text{risk} = 1 - 0.95012^{\exp(0.2459)} = 6.32\%$, which matches the test fixture exactly.
- **SCORE2 + SCORE2-OP coefficients** (sex-specific Cox terms, baseline survivals 0.9605/0.9776/0.7576/0.8082, and regional recalibration constants scale1/scale2 for all 4 regions $\times$ 2 sexes $\times$ 2 age bands) were sourced from the SCORE2 supplementary material, as transcribed in the open-source RiskScorescvd R package (github.com/dvicencio/RiskScorescvd, file 11_SCORE2_func.R). We then verified the implementation against the SCORE2 paper’s own abstract, which states the 10-year risk for a 50-year-old smoker with SBP 140 mmHg, total chol 5.5 mmol/L, HDL 1.3 mmol/L: men 5.9% (low region) / 14.0% (very-high), women 4.2% / 13.7%. **Our engine reproduces all four values exactly.**
- **PREVENT coefficients** for the 10-year base model (23 features $\times$ 5 outcomes $\times$ 2 sexes = 230 betas, of which we use the 46 ASCVD-relevant ones) were extracted directly from the open-source preventr R package (github.com/martingmayer/preventr) by loading its sysdata.rda binary in R and exporting the coefficient matrix. The preventr author cites Khan 2024 Supplementary Table S25 as the source. We then verified the implementation against the published reference example: female age 50, SBP 160 (on BP-treatment), total chol 200 mg/dL, HDL 45 mg/dL, no statin, diabetic, non-smoker, eGFR 90, BMI 35 $\rightarrow$ expected 9.2%, our engine returns 9.20%, agreement at 4 decimal places.

In all three cases, **we wrote the engine ourselves in pure Python** (no R bridge, no compiled binary dependency, no API calls) — the above sources were used to obtain numerical values that we then encoded into our own modules with our own input validation, error handling, spline transformations, and explainability layers.

4. Heart Score normalisation

The single most consequential design choice in this engine is how to map three different 10-year risk percentages (with different outcome definitions) into one 0–100 number. We rejected three tempting shortcuts:

- **Average the percentages directly.** Wrong: a 10% Framingham CVD risk and a 10% PREVENT-ASCVD risk represent different underlying probabilities, so arithmetic-averaging hides the disagreement.
- **Take the maximum.** Conservative but loses information. A 30% Framingham + 5% PREVENT + 5% SCORE2 average would produce 30% under this rule, which over-weights one model.
- **Train a regression to map our three inputs to one number.** The brief forbids ML training, and we wouldn't have the labelled outcome data anyway.

The chosen approach maps each percentage to a 0-100 sub-score using *that algorithm's own published clinical risk-band thresholds* with linear interpolation inside each band, then takes a weighted average.

Algorithm	0-20	20-40	40-60	60-80	80-100
Framingham	<5%	5-10%	10-20%	20-30%	>30%
SCORE2	<5%	5-10%	10-20%	20-30%	>30%
PREVENT	<5%	5-7.5%	7.5-10%	10-20%	>20%

The bands for Framingham and SCORE2 are the 5/10/20/30% breakpoints used clinically for low / borderline / moderate / high / very-high risk classification. PREVENT's bands follow the AHA/ACC 2026 lipid guidelines, which set 5% / 7.5% / 10% / 20% as the meaningful thresholds for lipid-lowering therapy decisions.

This approach has three properties we like: 1. We don't pretend to out-statistics the algorithms; we use each one's own calibration. 2. The bands are clinically validated decision thresholds, not arbitrary numbers. 3. The mapping is monotone and continuous, so small input changes produce small score changes.

Sub-model weights: - Framingham: 0.35 - PREVENT: 0.35 - SCORE2: 0.30

PREVENT is slightly favoured for being the most recent (2023) and for incorporating kidney function and BMI. SCORE2 is mildly down-weighted because India is not in its original 4-region calibration and we therefore default to the "high-risk" region as a proxy (see §6 below). Framingham remains weighted alongside PREVENT because it is the most thoroughly validated of the three across decades and populations, even though it is the oldest.

The final 0-100 score maps to a five-tier risk band:

Score	Band
0-19	Low
20-39	Borderline
40-59	Moderate
60-79	High
80-100	Very High

5. Explainability layer (top contributing factors)

For each sub-model we compute the per-factor contribution to the linear predictor, evaluated as the deviation from a matched “optimal reference patient” of the same age and sex (TC 170 mg/dL, HDL 60 mg/dL, SBP 110 mmHg, BMI 25, eGFR 90, no diabetes/smoking/medication). We then:

1. Normalise each model’s contributions by the sum of absolute contributions (so a small-coefficient model can’t be drowned out by a large-coefficient model).
2. Sum across models, harmonising names where the same underlying factor appears under different labels (e.g. Framingham’s “Total cholesterol” and PREVENT’s “Non-HDL cholesterol” both roll up into “Cholesterol (lipids)”).
3. Rank by descending contribution and return the top three positive drivers.

Output: a ranked list of human-readable factor names with relative weights, surfaced in the UI and in the JSON export.

6. Missing-data handling

The brief required handling a profile with missing cholesterol. Our policy:

- **Framingham** runs only if age 30–74 and lipid + SBP inputs are present. Lipid imputation is not attempted — Framingham *requires* TC and HDL.
- **SCORE2 / SCORE2-OP** runs only if age 40–89.
- **PREVENT** requires BMI and eGFR. If either is missing, PREVENT is skipped.

When any sub-model is skipped, the surviving sub-models are re-weighted proportionally so the weights still sum to 1. The skipped models are flagged in the result notes and shown in the UI so the user can see exactly what fed into the score and what didn’t. This is preferable to silent imputation.

7. Indian-population considerations and limitations

A few honest caveats:

- **SCORE2 has no India-specific calibration.** SCORE2 publishes recalibration constants for four European risk regions only. India’s cardiovascular epidemiology more closely resembles the “high” region (CV mortality between 150–300 per 100,000, prevalence of premature ASCVD comparable to Eastern Europe), so we default the SCORE2 region selector to “high” — but the user can override this. A SCORE2 Asia-Pacific variant exists (Hageman 2024) but is not yet implemented here; this is a clear next-iteration improvement.
- **PREVENT was developed and validated in US data.** The AHA explicitly flags this in its FAQ. PREVENT may over- or under-estimate risk in South Asian populations, who are known to have higher ASCVD risk at any given LDL level than the populations PREVENT was trained on.

- **Framingham notoriously over-estimates** in non-Anglo-Saxon populations. Several studies (e.g., D’Agostino 2001 internal validation; subsequent Indian validation studies) have shown it can over-predict by 30–60% in South Asian cohorts. Recalibration to Indian data is a known need.
- **None of the three include South-Asian-specific risk enhancers** (e.g. coronary calcium imaging, family history of premature CVD, Lipoprotein(a)). The 2019 ACC/AHA primary prevention guideline lists South Asian ancestry itself as a risk-enhancing factor, which our current engine does not surface.

The unified Heart Score is therefore best interpreted as a *triangulated educational estimate*, not a clinical decision-making tool. The risk band should be considered indicative, particularly for Indian patients. Any real CardioQ.ai product release would need (a) recalibration on Indian cohort data, (b) integration of South-Asian-specific enhancers, and (c) regulatory pathway compliance.

8. What I would build next given two more weeks

1. **Indian recalibration.** Re-fit the regional constants of SCORE2 to Indian cohort data (e.g. INTERHEART India arm, LUCAS, or similar) so the engine produces locally-calibrated risk rather than European “high-region” approximation.
2. **South Asian risk enhancers.** Add Lp(a), family history, and either CAC or CIMT imaging as optional inputs, with a documented uplift on the unified score.
3. **Lifetime risk view (PREVENT 30-year).** PREVENT’s 30-year equations are already in the same supplementary material; surfacing them would enable the “what does my risk look like at 65?” conversation that younger patients find more motivating than 10-year numbers.
4. **EHR integration via FHIR.** Plumb in an ingestion path that accepts a FHIR Bundle and extracts the 10-12 inputs automatically.
5. **Auditability.** Store all calculations with a hash of the inputs, the engine version, and the reference papers used, so a clinician can later reproduce what the patient saw on a given date.
6. **BMI/weight-loss as a quantified counterfactual.** Currently we don’t surface a BMI scenario because BMI isn’t a predictor in any of the three sub-models for the ASCVD outcome (it only enters PREVENT’s separate HF model). A defensible enhancement would be to model weight loss as also raising HDL and lowering SBP using published effect sizes (e.g. ~4 mmHg SBP per 5kg lost), then run the counterfactual through the same engine — but this requires its own validation and explicit user-facing caveats.

Note: counterfactual “what if” scenarios for smoking, blood pressure, HDL, and statin initiation are already implemented in this prototype (see `counterfactual_scenarios()` in `heart_score.py` and the “What if this patient changed something?” panel in the UI).